



PLANT REPRODUCTION — FULL MIND MAP

Biology Ch. 8.iv · Plant Physiology · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this sub-chapter:

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1. Meiosis in Flower & Reproductive Parts

- Plants mein meiosis study karne ke liye anthers (परागकोश) best part hain. Meiosis ke baad pollen grains bante hain, jo immature male gametophyte hote hain; stamens male reproductive part aur carpels female reproductive part hote hain. [Q1]

2. Vegetative / Clonal Propagation — Basics, Gootee & Clonal Selection

- Vegetative/clonal propagation asexual reproduction hai: bulbs, corms, tubers, cuttings jaise vegetative propagules successive mitosis se genetically similar clonal population banate hain. Ye most of the year practice ho sakta hai; source ke mutabik viruses eliminate nahi hote, balki vegetative parts se transmit bhi ho sakte hain. [Q2]
- Gootee = air layering / marcotting: parent plant se connected aerial stem par rooting karwa kar vegetative propagation ki jati hai. [Q3]
- Clonal selection asexually propagated crops ke selection ka method hai; selected plants se naye clones banaye jate hain. Source examples: potato, sugarcane aur banana. [Q8]
- Vegetative propagation mein offspring ek parent se banta hai aur genetically parent jaisa hota hai; root, stem ya leaf jaise vegetative parts se naya plant develop ho sakta hai. Stem cutting source ka common artificial example hai. [Q12]

EXAM TRAP — Vegetative Propagation & Virus

- ▶ Vegetative propagation → clonal population + most of the year possible. [Q2]
- ▶ “Vegetative propagation virus ko eliminate karta hai” source ke according FALSE; viral disease vegetative parts se transfer ho sakti hai. [Q2]

3. Sugarcane Propagation, Photoperiodism & Crop Rotation

- Sugarcane bud-chip settlings nursery mein raise karke field mein transplant karne se seed material ki saving hoti hai. Direct planting mein source ke mutabik many-budded setts ka germination single-budded se better, aur bad weather mein larger setts ki survival better hoti hai; tissue-culture settlings bhi use ho sakte hain. [Q4]
- Photoperiodism ke context mein sugarcane ko source short-day plant batata hai; winter/shorter days mein flowering se link kiya gaya hai, saath hi quantitative short-day/intermediate daylength behavior note kiya gaya hai. [Q5]
- Crop rotation mein deep-rooted crop ke baad shallow-rooted crop rakhna chahiye; legume ke baad non-legume useful hai kyunki fixed nitrogen ka effective use ho sakta hai. [Q6]

- Cereals ke saath pulses intercropping mein complementary crops act karte hain. Source A/R mein dono statements true hain: pulses ka tap-root system cereals ke fibrous roots se different depth use karta hai, lekin Reason ko Assertion ka complete/correct explanation nahi maana gaya. [Q7]
- Sugarcane predominantly stem cuttings se propagate hota hai; source contrast mein mango/cotton generally seeds se aur banana suckers se grow kiye jate hain. [Q13]

EXAM TRAP — Sugarcane Setts

- ▶ Bud-chip nursery settlings → seed material ki substantial saving. [Q4]
- ▶ Direct setts: many-budded germination better; bad weather mein large setts survival better — source statements 2 & 3 ko incorrect mark karta hai. [Q4]

4. Buds, Stem Cutting, Grafting & Layering

- Plant bud ek undeveloped/embryonic shoot hai; usually leaf axil ya stem/branch tip par milta hai. Axillary aur terminal buds ke form mein present hota hai. [Q9, Q16]
- Stem cuttings se source examples mein Bougainvillea, Carnation, Cocoa aur Grapes sab propagate kiye ja sakte hain; woody shrubs/ornamental plants mein ye popular method hai. [Q14]
- Sweet orange grafting se aur Jasmine layering se propagate kiya jata hai. Layering mein stem/branch ka part soil mein bury karke roots/shoots develop karwaye jate hain; grafting mein do plants ke tissues join karwaye jate hain. [Q15]

5. Flower Life Cycle — Embryo, Fruit, Seed, Wood & Starch

- Embryo seed ke andar milta hai; source ke mutabik embryo aur mother-plant derived other cells milkar next generation ka seed banate hain, jo germination ke baad new plant develop karta hai. [Q10]
- Source matching: Fruit → Ovary | Seed → Ovule | Wood → Stem (secondary xylem) | Starch → Leaf. [Q11]
- Plant life cycle ke terms mein flower ko source most important part batata hai: male-female gametes ki union mediate karta hai; sequence pollination → fertilization → seed formation → seed dispersal tak jata hai. [Q17]

6. Seed Germination — Water, Temperature & Oxygen

- Seed germination ke liye source ke according sabse necessary conditions hain: water + proper temperature + oxygen. Sunlight essential nahi maana gaya. [Q18, Q19]

EXAM TRAP — Germination Requirement

- ▶ Water + suitable temperature + oxygen = essential conditions. [Q18]
- ▶ Light/sunlight = generally NOT required in source question framing. [Q19]

7. Seed Dispersal — Censer Mechanism in Poppy

- Seed dispersal parent plant se seeds ko door le jane ki process hai. Censer mechanism poppy mein milta hai: wind se capsule sway karta hai aur seeds censer ki tarah shake out hote hain. [Q20]

8. Pollination — Geitonogamy, Agents & Dichogamy

- Same plant ke ek flower ke anther se doosre flower ke stigma tak pollen transfer = Geitonogamy. Functionally pollinating agent involve hone ki wajah se cross-pollination jaisa, lekin genetically same plant hone se autogamy jaisa hai. [Q21]
- Pollination = pollen grains ka male anther se female stigma tak transfer; fertilization ke liye prerequisite hai. Source pollinating agents mein bat, bee aur bird — teeno ko include karta hai; bees ko particularly important bataya gaya hai. [Q22, Q23]

- Bisexual flower mein androecium aur gynoecium different times par mature hon to Dichogamy kehte hain; self-pollination reduce hota hai. Types: Protandry (male first) aur Protogyny (female first). [Q26]



EXAM TRAP — Pollination Terms

- ▶ Geitonogamy = same plant ke different flowers ke beech pollen transfer. [Q21]
- ▶ Dichogamy = male & female reproductive parts ka different time par mature hona; Protandry / Protogyny. [Q26]

9. Hybrid Seed & Seed Multiplication Stages

- Har saal change kiya jane wala seed source ke according Hybrid seed hai, kyunki heterosis/hybrid vigour F1 generation mein strongest hota hai aur saved hybrid seed ki next generation segregate kar sakti hai. [Q24]
- Seed multiplication ka source chronological order: Nucleus seed → Breeder seed → Foundation seed → Certified seed. [Q25]

10. Crop Cytogenetics — Wheat Genome & Maize Endosperm

- Commercial edible/bread wheat (*Triticum aestivum*) mein source ABD genome (AABBDD) batata hai; ise allohexaploid species explain kiya gaya hai. [Q27]
- Maize mein $2n = 20$, gamete $n = 10$; angiosperm endosperm triple fusion se triploid ($3n$) hota hai, isliye maize endosperm mein 30 chromosomes hote hain. [Q28]

11. Flowering Stimulus — Role of Leaves

- Flowering stimulus receive karne wala plant component source ke according Leaves hain. Source explanation light/energy/ CO_2 reception aur florigen production in leaves ko shoot apex tak transport hone se link karta hai. [Q29]